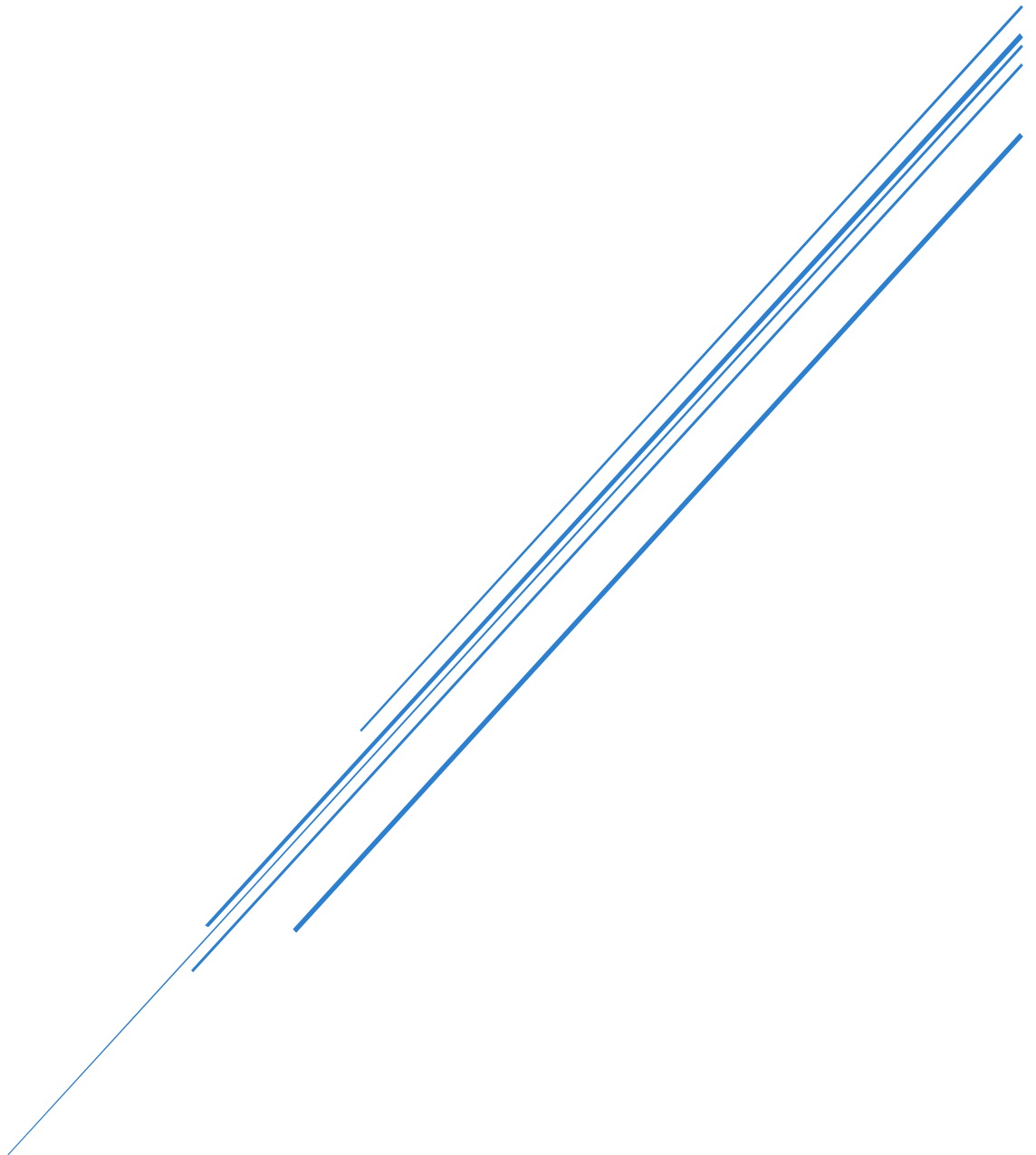


LAB 6 ANSWERS REPORT

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Artificial Intelligence (COMP3742)

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Question 1

Use the Open-Meteo API to fetch Adelaide's daily temperature data for Winter 2026 (June, July, August) and plot the time series. Identify any obvious trends or seasonality in the data. This data should be utilised for question 2-5 below.

As seen bellow in Figure 1: Decomposed Plot we can see that the temp ranges between near 0 and low 20's. The trend line starts and ends around the same spot, no monotonic increase or decrease, somewhat expected for weather data over just winter months.

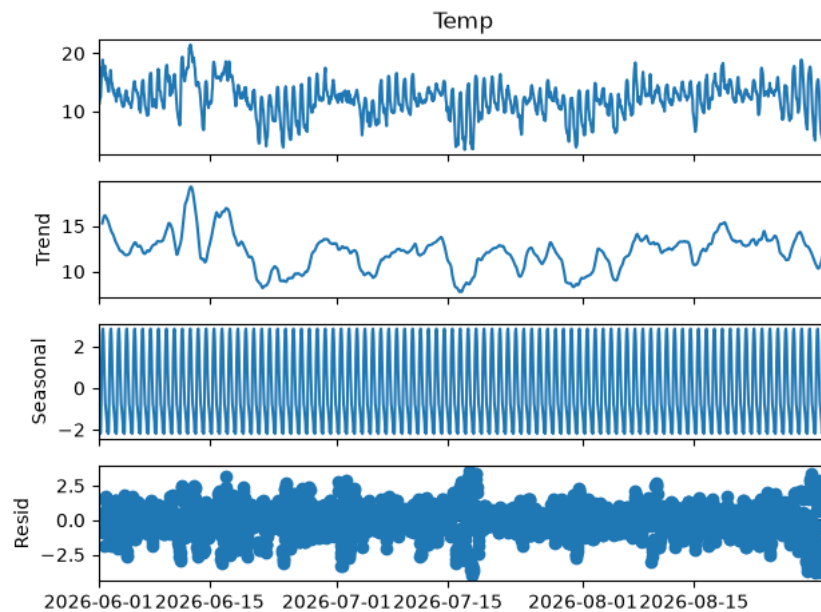


Figure 1: Decomposed Plot

Question 2

Fit an ARIMA model to the Adelaide daily temperature data for Winter 2026. Use the order (5,1,0) and calculate the training RMSE.

The RMSE = 0.5833617655583845 for the ARIMA model with order (5, 1, 0).

Question 3

Using the Adelaide daily temperature data for Winter 2026, compare the performance of ARIMA models with different orders: (5,1,0), (1,1,1), and (2,1,2). Calculate and compare the training RMSE for each model.

RMSE for order (5, 1, 0): 0.5833617655583845

RMSE for order (1, 1, 1): 0.5932028581823455

RMSE for order (2, 1, 2): 0.5852650751980312

We can see that there isn't much difference in the RMSE values for any of our orderings all are very close but (5, 1, 0) has the lowest error.

Question 4

Perform a 7-step rolling forecast on a hold-out test set using the best ARIMA model from the previous question. Calculate the MAE, MSE, RMSE, and R^2 for the rolling forecast.

For the 7-step rolling forecasts the error metrics were:

- Mean Absolute Error: 0.5563769483754284
- Mean Squared Error: 0.3745561033391023
- Root Mean Squared Error: 0.6120098882690559
- R^2 Score: 0.7836722175434228

It made our RMSE a tiny bit worse than before but still not too bad for our test set error values considering how close the RMSE is to its training set error.

Question 5

Update the model to include an exogenous variable (e.g., hour of the day). Fit the model and compare its performance with the best Adelaide daily temperature ARIMA model without the exogenous variable.

No Exogenous Variable:

- Mean Absolute Error: 0.40847331407517923
- Mean Squared Error: 0.3403109495153955
- Root Mean Squared Error: 0.40847331407517923

With Exogenous Variable:

- Mean Absolute Error: 0.408626540351655
- Mean Squared Error: 0.33984426183295185
- Root Mean Squared Error: 0.408626540351655

The exogenous variable makes near no difference only slightly decreasing the MSE and slightly increasing both MAE and RMSE. This may be because the exogenous variable was just hours as an integer, maybe if we used something more periodic it could help.